



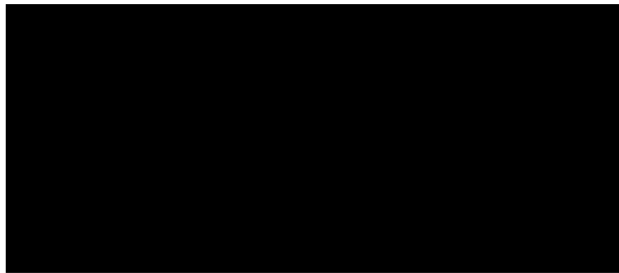
# INDIVIDUAL ASSIGNMENT PART 1

TECHNOLOGY PARK MALAYSIA

CT015-3-2

DESIGN METHODS

APD2F2409SE



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## **1.0 Project Background**

As technology advances, it makes people more efficient, easier, faster, and more organized in the context of project management. It helps users plan, track, and manage their tasks, schedules, deadlines, and team collaboration. Notion is primarily a planner management application and cloud-based productivity and collaboration tool. It's designed to be a single space for capturing thoughts, managing projects, and organizing information, whether for personal or professional use (Lau, 2024). In 2016, Notion was released as the first generation of desktop on San Francisco, and the android app released in 2018. Notion is one of the best planner systems in the world. It provides many useful functions to let the users easily to manage their project and easily to share and access with the different area users to complete the task with good collaboration and communication on the teamwork. Furthermore, it also can save time using useful functional features to create content, take notes, organize projects, automate workflows, or assign task (Deer & Nevena Radulović, 2023). According to Ivan Zhao, Notion has passed 100 million users in September 2023. It is more than its competitor, Trello.

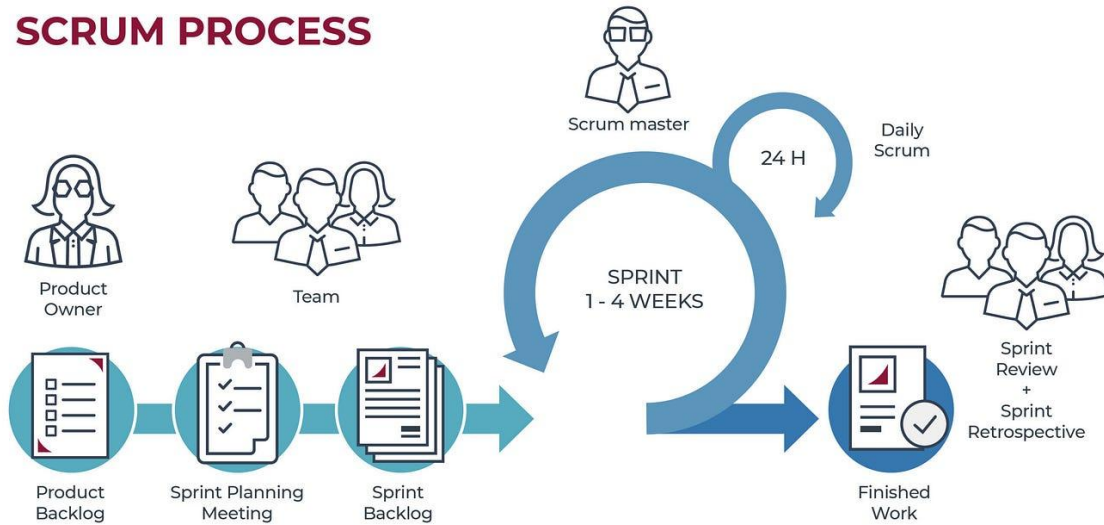
However, despite its comprehensive functionality, Notion faces challenges, particularly when handling complex projects. After users entry their tasks and schedules, they are not easy to clearly understand tasks and projects progress when dealing with complex pages and multiple nested documents. Raymond was said in 2023, this inefficiency can significantly impact productivity, especially for users managing multiple projects or collaborating with large teams. While using the visualize graph in project dashboards can ensure the user clearly visualizes overall tasks progress and let users make decisions efficiently with easy management. By using the Application Programming Interface (API) to connect the frontend graph components and backend database in real time analysis data. This also can improve the system without user manual updates or data entry.

Moreover, Notion lacks a dedicated and well-integrated time tracking feature compared to Trello. Trello is offering third-party time tracking plugins and power-ups that seamlessly record task durations and deadlines, Notion still relies heavily on manual input or external integrations for time management. This shortcoming limits its usefulness for teams that require precise time allocation, performance tracking, or automatic deadline reminders as part of their project planning workflow. By the external system and application to achieve the real time tracker is Clockify. Clockify is a web-based time tracker and timesheet app that lets you track work hours across projects. Notion can use the API to connect the Clockify data and real time tracker system's data to database and show the result of each task and projects. The time tracker also needs to link the hardware, software, and system which the users open or start their work. This can let the users to improve their time management and productivity for eliminating time wasting activities and accurate project estimation for setting the projects and tasks dateline

(Sorokina et al., 2021).

Hence, this project aims to improve the planner management system which is Notion by developing visualize graph functions in projects overall dashboard and the time tracking function to provide more efficiency, easy management fast navigation and efficient time tracking in Notion Application.

## 2.0 Software Development Methods



*Figure 1 Scrum Phases Diagram*

A software development methodology is a process by which developers design, implement and test new computer programs. Following a methodology benefits developers because these processes lay out a structured sequence of steps that guide professionals through each stage of development. It has many methods to consider the project such as Agile, Scrum, Waterfall, and Spiral. In these methodologies, Scrum Methodology is the suitable methodology to develop the Notion application. Scrum is a lightweight Agile framework that focuses on delivering functional software through iterative and incremental development cycles called “Sprints.” Each Sprint typically for one to four weeks and results in a potentially shippable product increment, allowing teams to deliver value quickly and adapt to evolving requirements (Artem Gurnov, 2022). It emphasizes continuous feedback, collaboration, and transparency, making it highly suitable for this project, which involves developing a visualization graph, an overall dashboard, and an integrated time tracking function. Each Sprint will also include planning the development of these features, enabling the team to regularly collect user feedback and iterate on features based on real user needs, usability testing, etc. Moreover, Scrum promotes the roles of the benefited owner, scrum master, and development team thus facilitating clarity about roles and the consequent efficient working together. This structure will help reduce user feedback loops, catch detections at an early stage, and improve the front end and back end in accordance with actual use cases. Scrum’s responsiveness and focus on the user have made it the perfect technique to ensure the final Notion enhancement is as solid yet easy to use as possible.

### 3.0 Use Case Diagram



Figure 2 Planner Management Application Notion's Use Case Diagram

## **4.0 Use Case Specifications**

### **1. UC1\_Create Project based with Templates or Tools**

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Create project based with templates or tools                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description      | Planner can create a new project dashboard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner successfully created a new project dashboard</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Standard Process | <ol style="list-style-type: none"> <li>Planner clicks 'Create New Page' button.</li> <li>System displays an empty dashboard.</li> <li>Planner clicks the "Template" button on side bar</li> <li>System loading templates list</li> <li>System displays templates list</li> <li>Planner selects the template</li> <li>System loads the selected project templates information</li> <li>System displays the selected project templates information</li> <li>Planner clicks the "Add" button to create a new project</li> <li>System downloads the template</li> <li>Planner clicks the project's selector bar on side bar</li> <li>System displays project dashboard</li> <li>Planner view projects dashboard</li> </ol> |
| Alternative Path | <p><b><u>3a Select Personalization Tools</u></b></p> <ol style="list-style-type: none"> <li>Planner selects the personalization tools.</li> <li>System download and saved selected tools.</li> <li>System displays the tools.</li> <li>System continues in step 12.</li> </ol> <p><b><u>3b. Not Select Any Templates and Tools</u></b></p> <ol style="list-style-type: none"> <li>Planner view the empty dashboard.</li> </ol>                                                                                                                                                                                                                                                                                         |
| Exception Flow   | <p><b><u>5a. Invalid Access Template</u></b></p> <ol style="list-style-type: none"> <li>System failed to access template.</li> <li>System display "We couldn't find that page" message.</li> </ol> <p><b><u>7a. Failed Download Template</u></b></p> <ol style="list-style-type: none"> <li>System cannot download template.</li> <li>System display "Please go online to copy this block" message.</li> <li>Planner clicks "OK" button.</li> </ol>                                                                                                                                                                                                                                                                    |



## 2. UC2\_Select Workspace to View Project on Sidebar

|                  |                                                                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Select Project Dashboard to View Project on sidebar                                                                                                                                                                                                                    |
| Actor            | Planner                                                                                                                                                                                                                                                                |
| Description      | For planner select the project's workspace which has already been created to view the project dashboard                                                                                                                                                                |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create a project dashboard</li> </ul>                                                                                                                             |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the project</li> </ul>                                                                                                                                                                                         |
| Standard Process | <ol style="list-style-type: none"> <li>System displays a home page and side bar.</li> <li>Planner selected the specific workspace on side bar</li> <li>System Display the project dashboard and tasks</li> <li>Planner view the project dashboard and tasks</li> </ol> |
| Alternative Path | -                                                                                                                                                                                                                                                                      |
| Exception Flow   | <b><u>2a. No Existing Project</u></b><br>2a.1 Planner need to create a project                                                                                                                                                                                         |

## 3. UC3\_Manage Task

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Task Manage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description      | For Planner to create, modify, or delete their task in a project dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must create a project dashboard</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the latest updated tasks in project dashboard</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                   |
| Standard Process | <ol style="list-style-type: none"> <li>System displays the project dashboard and task.</li> <li>Planner views the project dashboard and tasks.</li> <li>Planner clicks the "New Task" button to add a task.</li> <li>System displays the task information form.</li> <li>Planner completed to fill the form.</li> <li>Planner saved the task in project.</li> <li>System displays the task title and status on project dashboard.</li> <li>Planner views the dashboard and updated task</li> </ol> |
| Alternative Path | <b><u>3a. Modify Tasks Information</u></b><br>3a.1. Planner selects the task to modify the information.<br>3a.2. System displays task information forms.<br>3a.3. Planner continues in step 5.<br><br><b><u>3b. Delete Task</u></b>                                                                                                                                                                                                                                                                |

|                |                                                                                                                                                                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | 3b.1. Planner clicks more action button.<br>3b.2. Planner selects delete section to remove the task.<br>3b.3. System removes the task and moves it to trash.<br><br><b><u>3c. Have Not Any Motion</u></b><br>3c.1 Planner have not added, modified, and deleted task. |
| Exception Flow | -                                                                                                                                                                                                                                                                     |

#### 4. UC4\_Set Time & Date

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Set Time & Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description      | For planner to set the due date and time in the projects and tasks to release the timeline                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create a project dashboard</li> <li>Planner must create a new task or modify the tasks.</li> </ul>                                                                                                                                                                                                                                                                                                               |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner set or change the date and time in the project and time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| Standard Process | <ol style="list-style-type: none"> <li>System displays the task information form</li> <li>System retrieves and checks today's date</li> <li>System pre-filled with today's date in date selector</li> <li>System detects time selector</li> <li>System displays the calendar if the time selector is off</li> <li>Planner selects desired due date</li> <li>System records and saved selected due date</li> <li>System updated due date on associated projects and tasks</li> <li>Planner view the updated</li> </ol> |
| Alternative Path | <b><u>4a. Time Selector Opened</u></b><br>4a.1. Planner opens time selector<br>4a.2. System displays the calendar, time selector with initial time, time zone, and time format<br>4a.3. Planner selects the due date, due time, time zone, and time format<br>4a.4. System record and saved selected due date, due time, time zone, and time format.<br>4a.5. System updated due date and time on associated projects and tasks<br>4a.6. System resumes in Step 9                                                     |
| Exception Flow   | <b>7a. Invalid Date and Planner Cancel Updated Date</b><br>7a.1. Planner inputs an invalid date<br>7a.2. System displays an "Invalid Date" message<br>7a.3. Planner cancels the date update                                                                                                                                                                                                                                                                                                                           |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>7a.4. System reverts and pre-fills the previously updated date in the calendar</p> <p><b>7b. Invalid Date and Planner Re-Enter Valid Date</b></p> <p>7b.1. Planner inputs an invalid date</p> <p>7b.2. System displays “Invalid Date” message</p> <p>7b.3. Planner enters a valid date</p> <p>7b.4. System resumes at Step 7</p> <p><b>7c. Invalid Time and Planner Cancel Updated Time</b></p> <p>7c.1. Planner inputs an invalid time</p> <p>7c.2. System display “Invalid Time” message’</p> <p>7c.3. Planner cancels the time update</p> <p>7c.4. System reverts and pre-fills the previously updated time in the calendar</p> <p><b>7d. Invalid Time and Planner Re-Enter Valid Time</b></p> <p>7d.1. Planner inputs an invalid time</p> <p>7d.2. System displays “Invalid Time” message</p> <p>7d.3. Planner enters a valid time</p> <p>7d.4. System resumes at Step 7</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5. UC5\_Open Relevant File with URL Link

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Add relevant file with URL link                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description      | For planner upload their relevant Google File, Figma, and GitHub link in specific task to access and find the file quickly.                                                                                                                                                                                                                                                                                                                                                                                        |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create a project</li> <li>Planner must create a task.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner view associated file and websites with URL link in the task information form</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |
| Standard Process | <ol style="list-style-type: none"> <li>System displays the task.</li> <li>Planner presses embed google drive button.</li> <li>System displays the embed link selector bar.</li> <li>Planner views link selector bar.</li> <li>Planner inputs Google file link.</li> <li>Planner clicks link button.</li> <li>System records and saved the file link.</li> <li>System displays saved link in link selector bar .</li> <li>System verified link successfully.</li> <li>System accesses Google Drive File.</li> </ol> |

|                  |                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------|
|                  | 11. Planner click link selector bar within Drive Link.<br>12. System opens the browser to access the file.    |
| Alternative path | -                                                                                                             |
| Exception Flow   | <b><u>9a: System Failed Verified Link</u></b><br>9a.1. System display “Unrecognized URL” on link selector bar |

## 6. UC6\_Personal Notification & Reminder

|                  |                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Personal notification & reminder                                                                                                                                                                                                                                                                                                                                                      |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                               |
| Description      | For notifying the planner of comments, mentions, and replies by other planners                                                                                                                                                                                                                                                                                                        |
| Precondition     | Planner must sign in to the application                                                                                                                                                                                                                                                                                                                                               |
| Postcondition    | Planner receives notifications for relevant activities                                                                                                                                                                                                                                                                                                                                |
| Standard Process | <ol style="list-style-type: none"> <li>1. System monitors planner-related activities.</li> <li>2. System detects relevant notification trigger in new comments, mentions, and replies.</li> <li>3. System checks user id available.</li> <li>4. System generates notification content.</li> <li>5. System display notification.</li> <li>6. Planner receives notification.</li> </ol> |
| Alternative path | -                                                                                                                                                                                                                                                                                                                                                                                     |
| Exception Flow   | <b><u>3a. Planner ID Unavailable</u></b><br>3a.1. System detects errors.<br>3a.2. System generates “User Not Found” and suggestion messages<br>3a.3. System continue in step 6                                                                                                                                                                                                        |

## 7. UC7\_Solve Questions Using Notion AI

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Solve Questions Using Notion AI                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Description      | For planner to solve their questions by using the AI                                                                                                                                                                                                                                                                                                                                                                                                        |
| Precondition     | <ul style="list-style-type: none"> <li>• Planner must sign in system</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |
| Postcondition    | <ul style="list-style-type: none"> <li>• Planner have received the AI’s generated answers</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| Standard Process | <ol style="list-style-type: none"> <li>1. System displays AI chatbot, text area, and submits button</li> <li>2. Planner views the AI chatbot, text area, and submits button</li> <li>3. Planner input the question</li> <li>4. System received question</li> <li>5. Notion AI model process question</li> <li>6. Notion AI model generated answers</li> <li>7. System displays the Notion AI model’s generated answer to the planner in chatbot.</li> </ol> |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | 8. Planner view the answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Alternative Path | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Exception Flow   | <p><b><u>5a. Planner Inputs Empty Question.</u></b></p> <p>5a.1. System displays error messages</p> <p>5a.2. System prompts planner to input valid questions</p> <p>5a.3. Planner continues in Step 3.</p> <p><b><u>6a. Notion AI Retrying When Fails to Process Question</u></b></p> <p>6a.1. Notion AI retrying process.</p> <p>6a.2. Notion AI continues in Step 5.</p> <p><b><u>6b. System End When Fails to Process Question</u></b></p> <p>6b.1. Notion AI generates error and invalid message.</p> <p>6b.1. System resumes on Step 8</p> |

### 8. UC8\_Searching Project and Task

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Searching Project and Task                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Description      | For planner to search associated projects in database                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must create the project.</li> <li>Planner must sign in to application.</li> </ul>                                                                                                                                                                                                                                                                                                                                      |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the associated projects and tasks in a list</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| Standard Process | <ol style="list-style-type: none"> <li>Planner clicks the search bar button on side taskbar.</li> <li>System displays searching selector bar and all project in a list.</li> <li>Planner inputs the name of projects or tasks in the selector bar.</li> <li>System compares names of projects and tasks based on alphabets.</li> <li>System displays the projects and tasks which successfully compare.</li> <li>Planner views list of projects and tasks.</li> </ol> |
| Alternative Path | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Exception Flow   | <p><b><u>5a. Failed Matching</u></b></p> <p>5a.1 System display Not Result and AI action button</p> <p><b><u>6a. Real-Time Search Looping When Planner Continue Typing</u></b></p> <p>6a.1 System resumes to Step 4</p>                                                                                                                                                                                                                                               |

### 9. UC9\_Sharing Project Dashboard

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | Sharing Project Dashboard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description      | For planner to share project dashboard and its contents with other team members or stakeholders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create a project dashboard</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner successfully shared with other planners</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Standard Process | <ol style="list-style-type: none"> <li>Planner selected a project</li> <li>Planner clicks the "Share" button</li> <li>System displays sharing dialog with options</li> <li>Planner inputs recipient email addresses</li> <li>Planner sets appropriate access permissions in view or edit</li> <li>Planner clicks "Share" to confirm</li> <li>System validates sharing permissions</li> <li>System sends access notifications to recipients</li> <li>System displays "Success Send Invite" message</li> <li>System records and saved the other planner email</li> <li>System displays other planner email in dialog</li> <li>Other planner receives access notification</li> <li>Other planner can open the dashboard</li> </ol> |
| Alternative Path | <p><b><u>4a. Copy Link to Share Project Dashboard</u></b></p> <ol style="list-style-type: none"> <li>Planner wants use link to share the project dashboards.</li> <li>Planner clicks copy button.</li> <li>System generated URL link</li> <li>System display "Copied link to clipboard" message.</li> <li>Planner sets appropriate access permissions in view or edit</li> <li>Planner send to other planner</li> <li>Other planner receive the link</li> <li>Other planner click the link</li> <li>Other planner continue in step 13</li> </ol>                                                                                                                                                                                |
| Exception Flow   | <p><b><u>5a. Invalid Email</u></b></p> <ol style="list-style-type: none"> <li>Planner input invalid email</li> <li>System displays invalid email</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### 10. UC10\_View Trash History

|               |                                                                                            |
|---------------|--------------------------------------------------------------------------------------------|
| Use Case Name | View Trash History                                                                         |
| Actor         | Planner                                                                                    |
| Description   | For planner to view and manage items that have been moved to trash.                        |
| Precondition  | <ul style="list-style-type: none"> <li>Planner must sign in to the application.</li> </ul> |

|                  |                                                                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <ul style="list-style-type: none"> <li>Planner must create projects or task.</li> </ul>                                                                                                          |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the deleted projects and tasks in trash history.</li> </ul>                                                                              |
| Standard Process | <ol style="list-style-type: none"> <li>Planner selects trash selector bar</li> <li>System loading data</li> <li>System displays trash history</li> <li>Planner view the trash history</li> </ol> |
| Alternative Path | <b>4a. Filter Projects and Tasks in Trash</b><br>4a.1 Planner Select Filter Rules<br>4a.2 System Continues in Step 2                                                                             |
| Exception Flow   | <b>2a. No Exists Projects and Tasks in Trash</b><br>2a.1. System displays "Trash is empty" mention in Trash.                                                                                     |

### 11. UC11\_Using Real Time Tracker

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | View Real Time Tracker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Actor            | Planner, Clokify                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Description      | For Planner to tracker their real time using in the projects or tasks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create projects or tasks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the time taken on their projects or tasks</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Standard Process | <ol style="list-style-type: none"> <li>Planner opens the projects</li> <li>System displays the project dashboard and task</li> <li>Planner ticks the time tracker's box to open the tracker system</li> <li>System connects and accesses the third-party system as Clockify with using API.</li> <li>System successfully connects and accesses</li> <li>Planner open the hardware, software or system to work</li> <li>Clockify detects the planner start work</li> <li>Clockify starts record the time and take the attendance</li> <li>System recording and copy the Clockify data</li> <li>Planner end the close the hardware, software or system</li> <li>Clockify detects the planner end work</li> <li>Clockify ends to record the time and display the offline time</li> <li>System stores and processes final time logs, making them available for reports, billing, or audit.</li> <li>Planner view the real-time using in the project dashboard</li> </ol> |
| Alternative Path | <b><u>10a. Short Time No Motion</u></b><br>10a.1. Planner no motion on hardware, software or system<br>10a.2. System stops recording time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>10a.3. System starts the breaktime timer<br/> 10a.4. Planner back to use hardware, software, system<br/> 10a.5. System stops breaktime timer<br/> 10a.6. System continues to step 7.</p> <p><b><u>10b, Long Time No Motion</u></b><br/> 10a.1. Planner no motion on hardware, software or system<br/> 10a.2. System stops recording time<br/> 10a.3. System starts the breaktime timer<br/> 10a.4. System timer over break time<br/> 10a.5. System continues to step 12.</p> |
| Exception Flow | <p><b><u>4a. Failed Connect Clockify</u></b><br/> 4a.1. System failed to connect Clockify<br/> 4a.2. Planner try to restart application or network or check the time tracker has opened or not.</p> <p><b><u>9a. Failed Access Data</u></b><br/> 9a.1. System failed to connect Clockify Database or copy data<br/> 9a.2. Planner try to restart applications or network.</p>                                                                                                   |

## 12. UC12\_View Visualize Graph

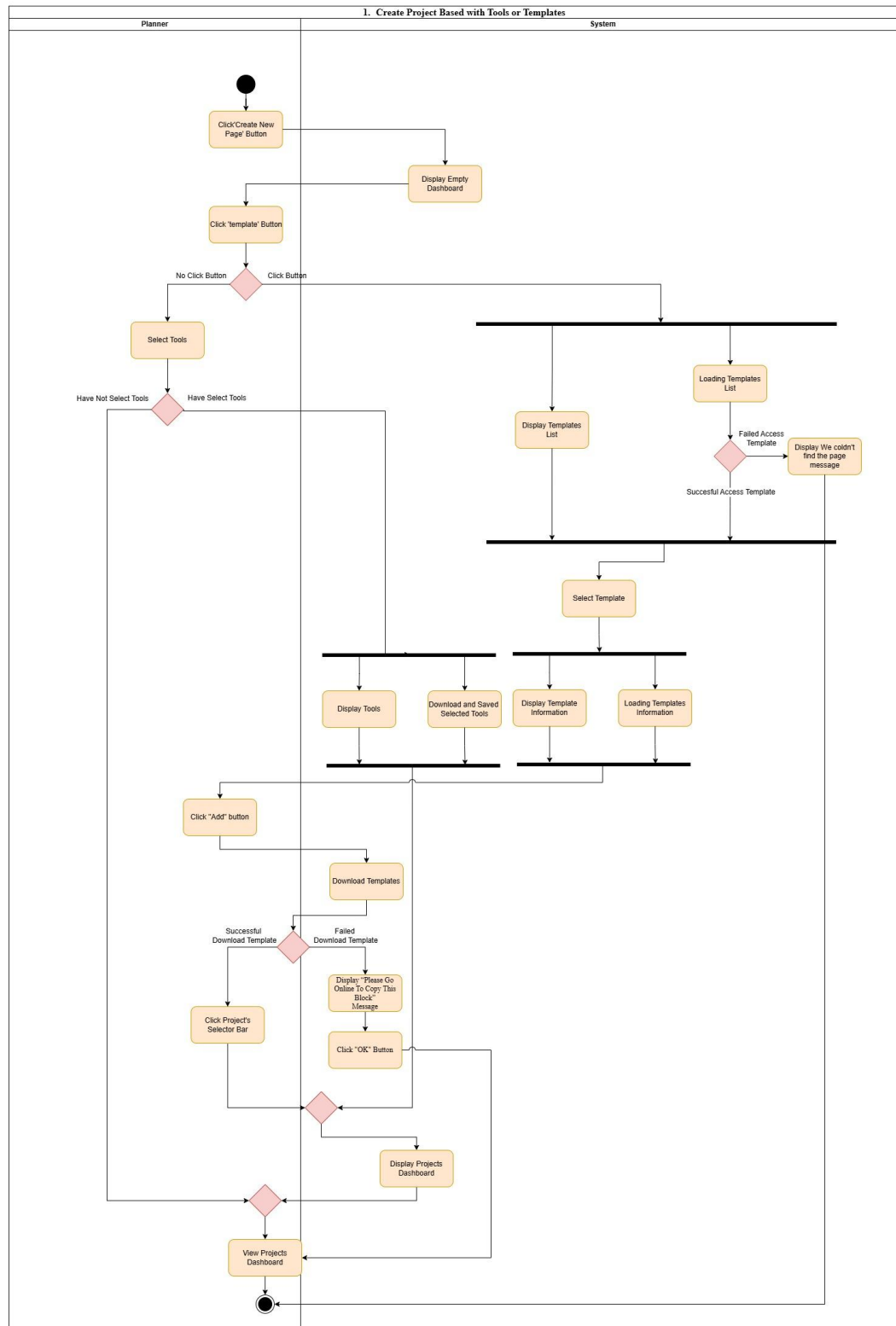
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name    | View Visualize Graph                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Actor            | Planner                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description      | For planner to view the overall status of projects and tasks in visualize graphs                                                                                                                                                                                                                                                                                                                                                        |
| Precondition     | <ul style="list-style-type: none"> <li>Planner must sign in to the application</li> <li>Planner must create the projects</li> </ul>                                                                                                                                                                                                                                                                                                     |
| Postcondition    | <ul style="list-style-type: none"> <li>Planner can view the visualize graph</li> </ul>                                                                                                                                                                                                                                                                                                                                                  |
| Main Flow        | <ol style="list-style-type: none"> <li>Planner opens the project dashboard</li> <li>System displays the project dashboard.</li> <li>System frontend request data by using API</li> <li>System loading and analysis overall of the project and tasks</li> <li>System generates visualize graph</li> <li>System embeds visualize graph on project dashboard</li> <li>Planner can view the visualize graph on project dashboard</li> </ol> |
| Alternative path | -                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Exception Flow   | <p><b><u>3a. Failed to Connect API</u></b><br/> 3a.1 System failed to connect API<br/> 3a.2. System records the error details</p>                                                                                                                                                                                                                                                                                                       |

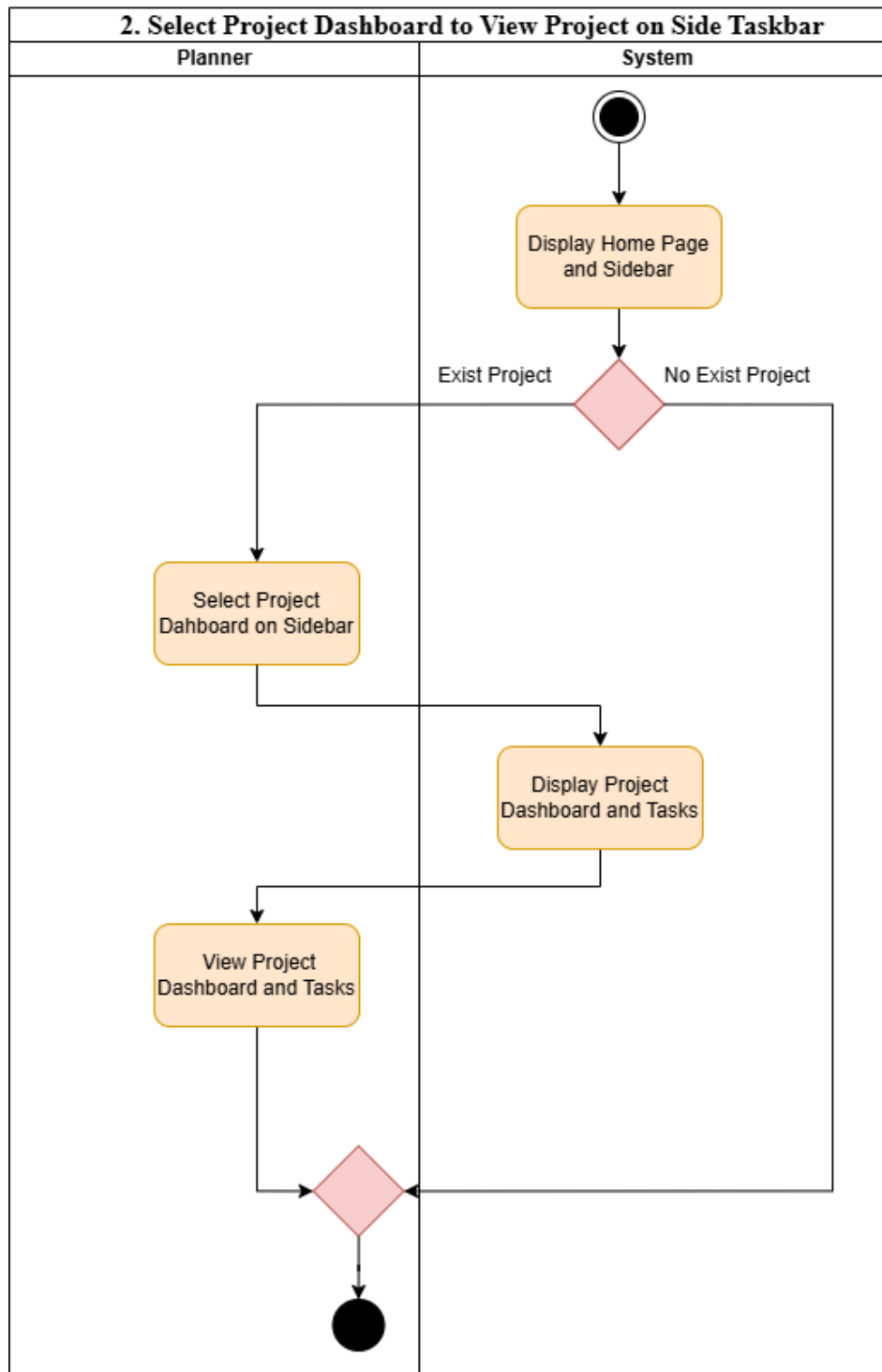


|  |                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>3a.3. System displays "Failed to load project data" message.</p> <p>3a.4. Planner close the application or check network to try again</p> <p><b><u>4a. Failed to Access Data</u></b></p> <p>4a.1. System failed to access the data</p> <p>4a.2. System record data error quantity and reasons</p> <p>4a.3. System creates a list or error data</p> <p>4a.4. System display the error of list data</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

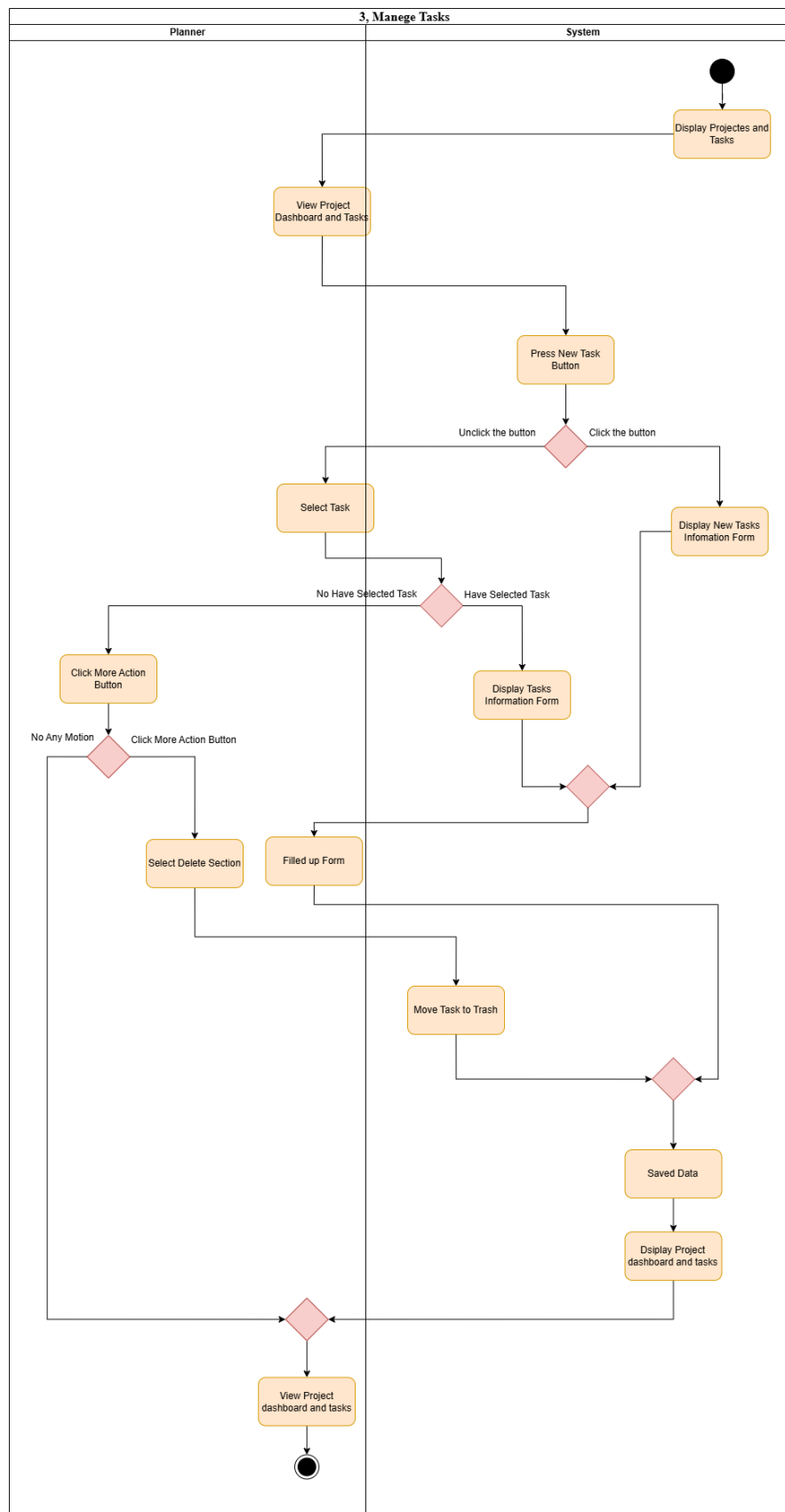
## 5.0 Activity Diagram

### 1. Create Project based with Tools or Templates

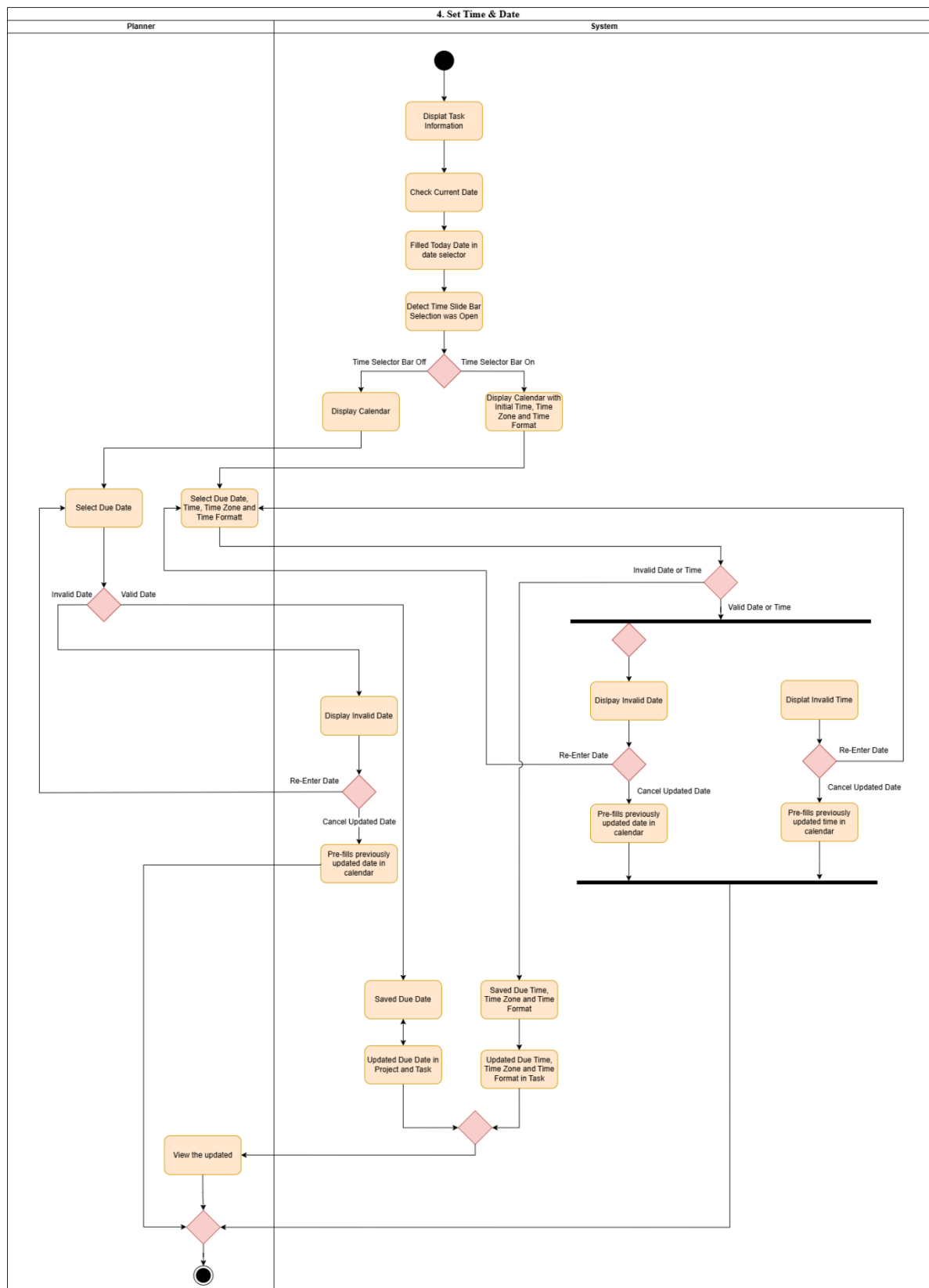


**2. Select Workspace to View Project on Side Taskbar**

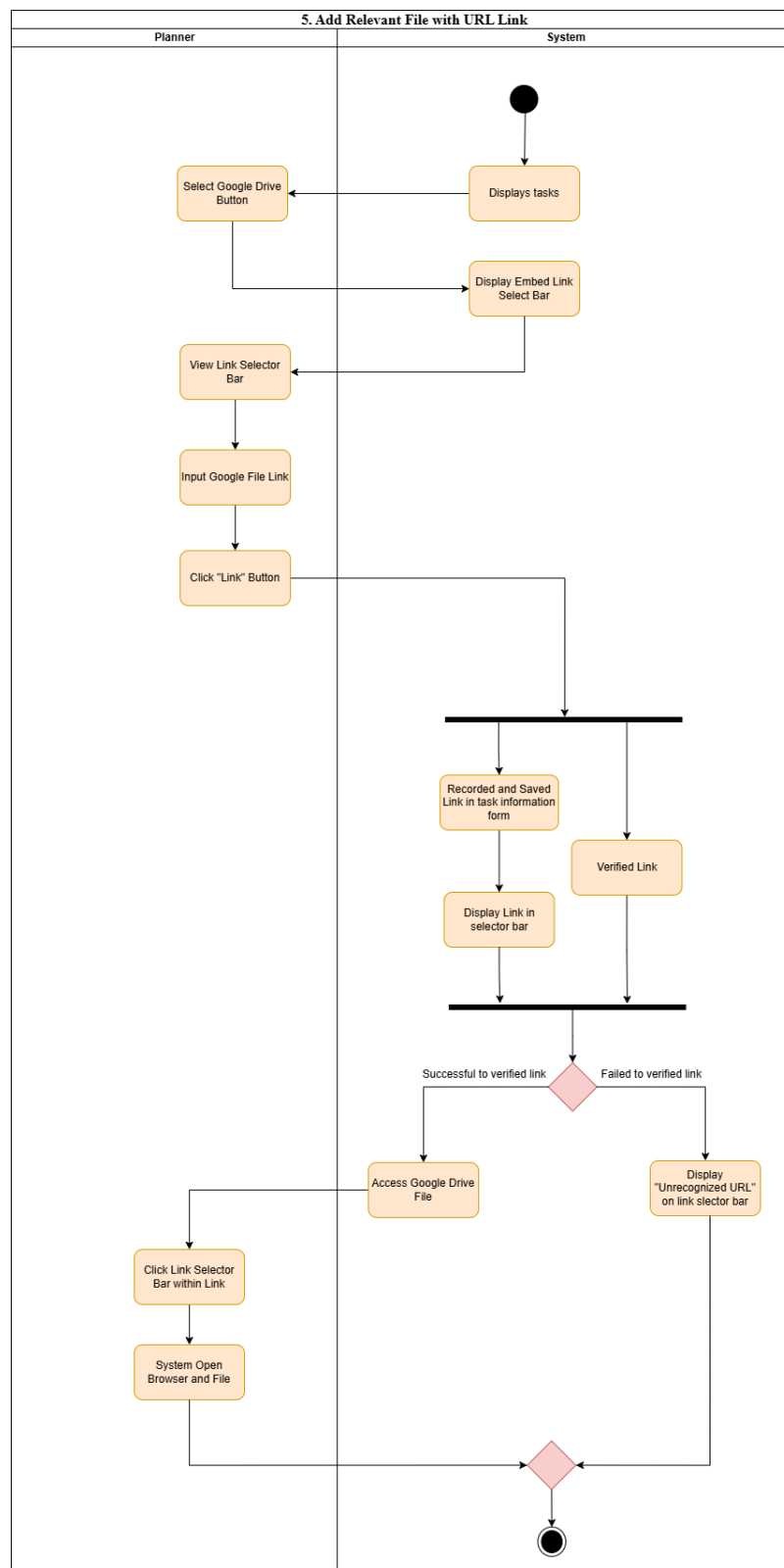
### 3. Manage Task



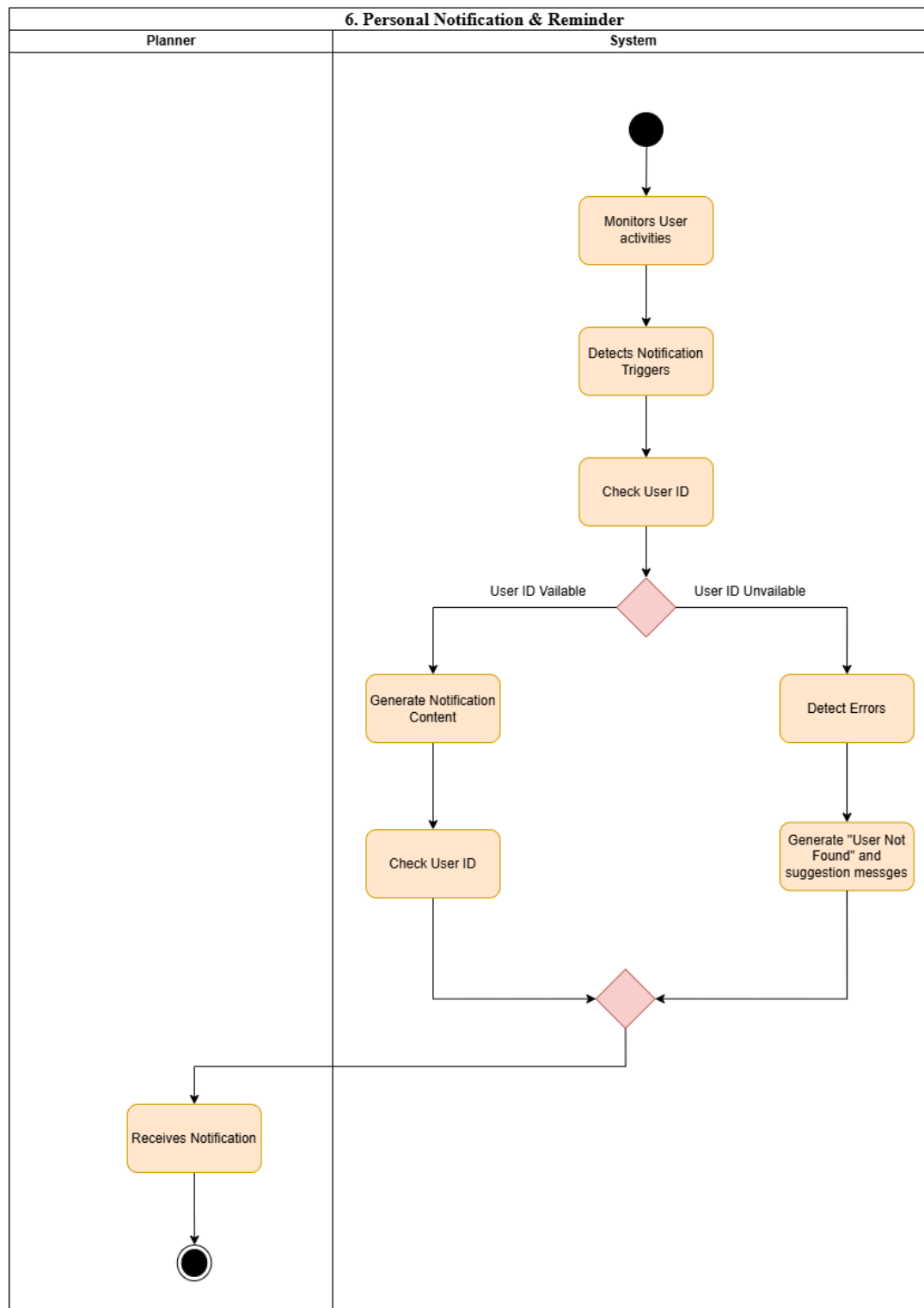
## 4. Set Time & Date



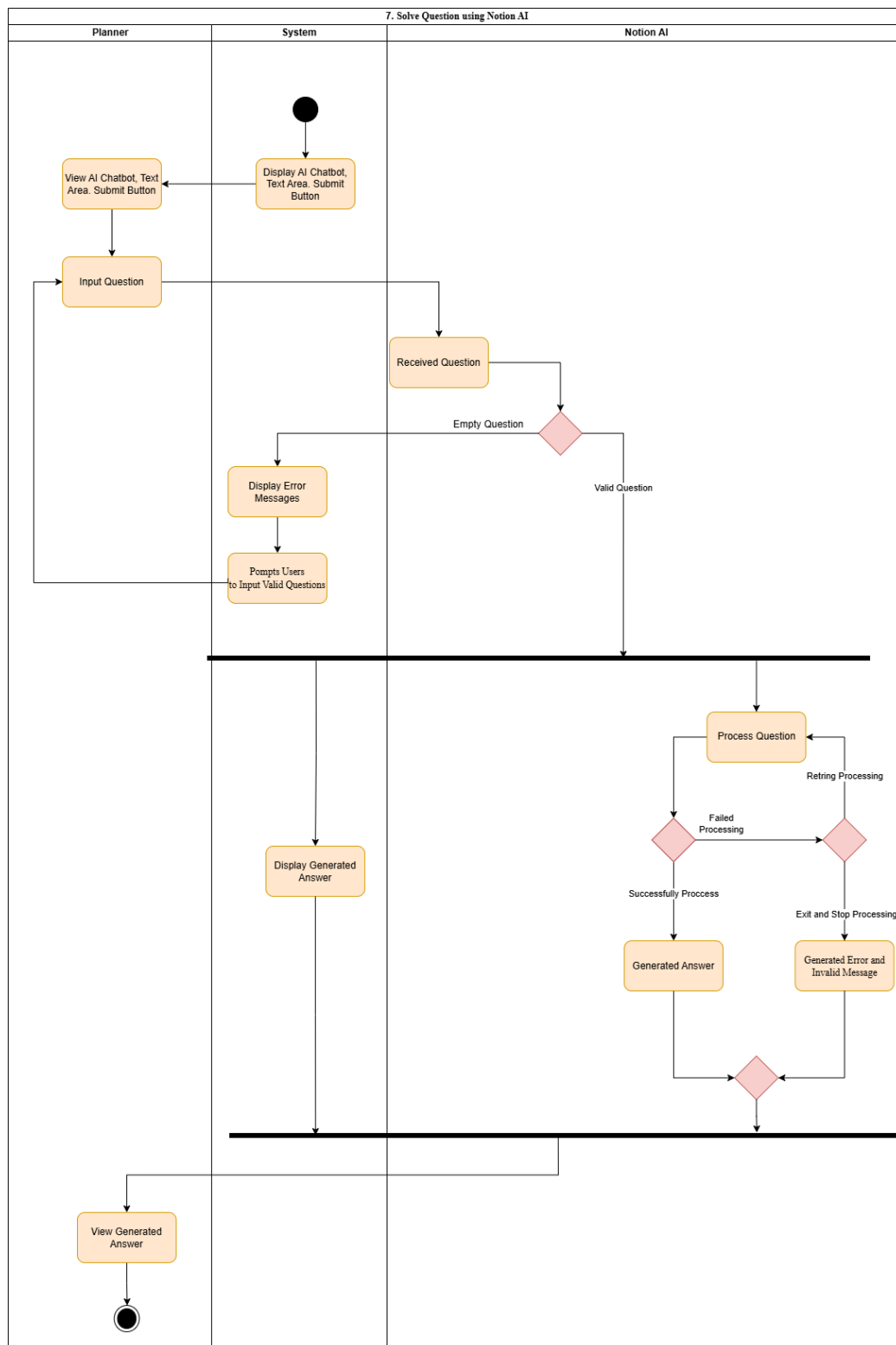
## 5. Add Relevant File with URL Link



## 6. Personal Notification and Reminder

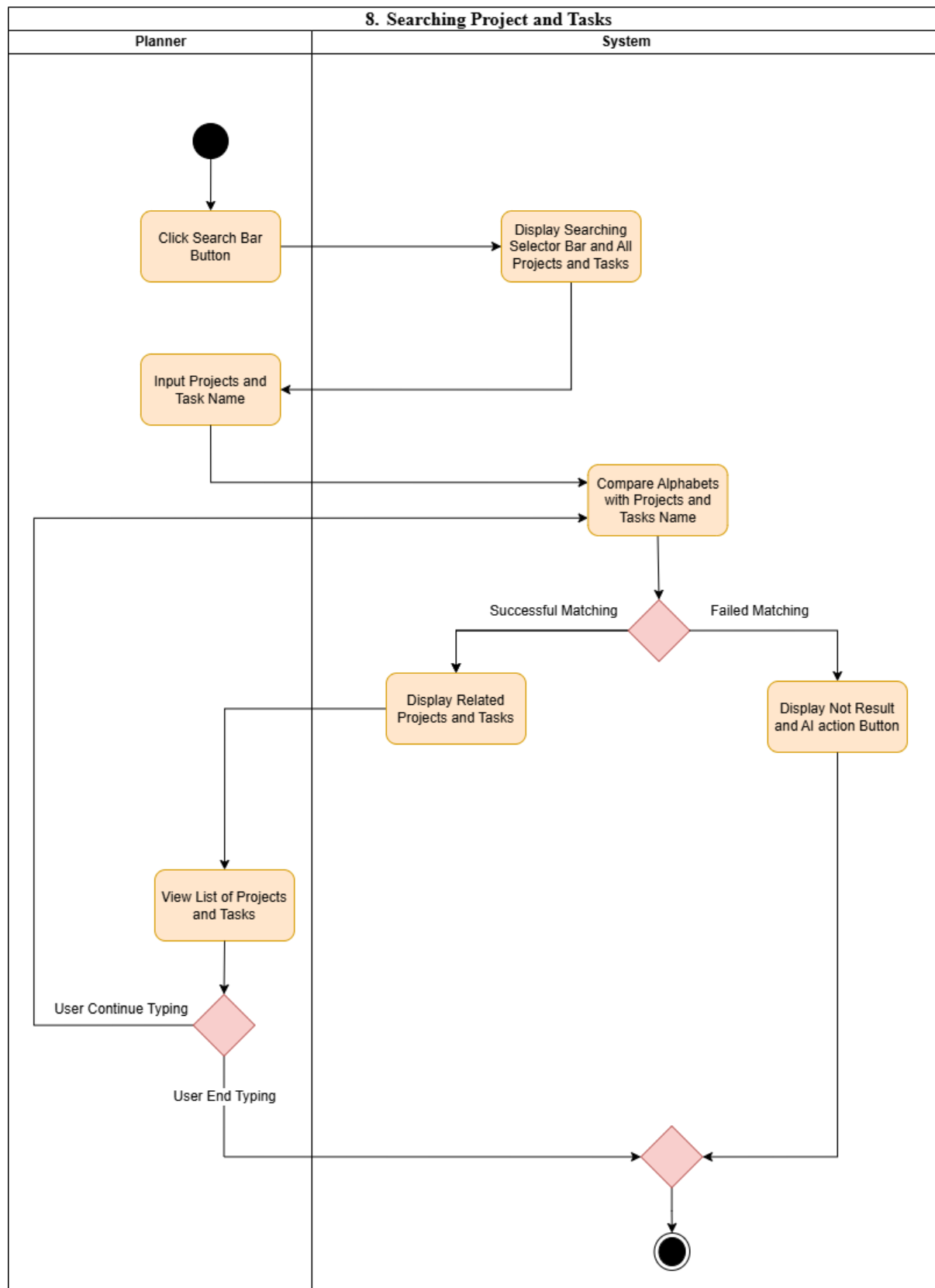


## 7. Solve Questions Using Notion AI

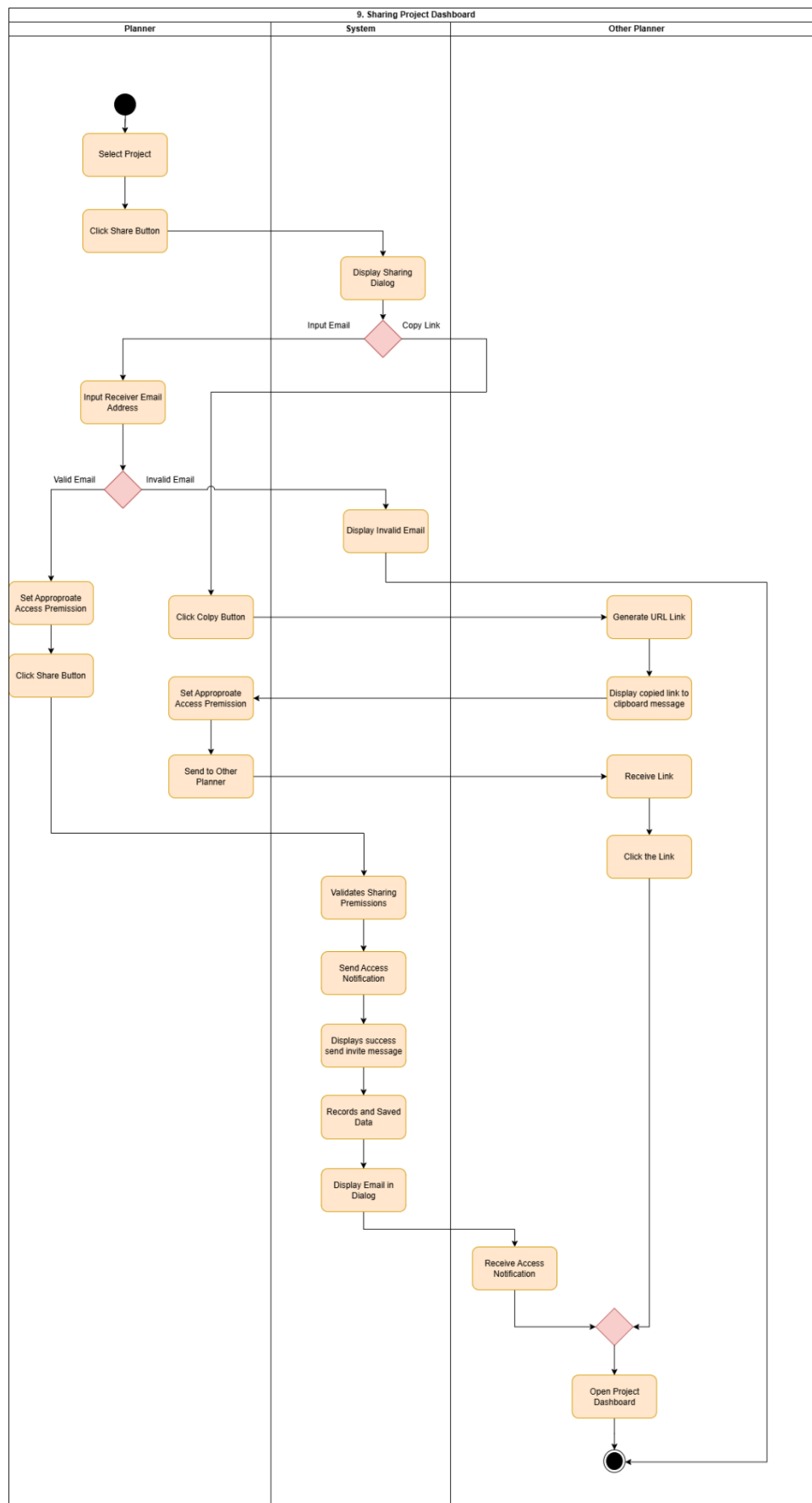


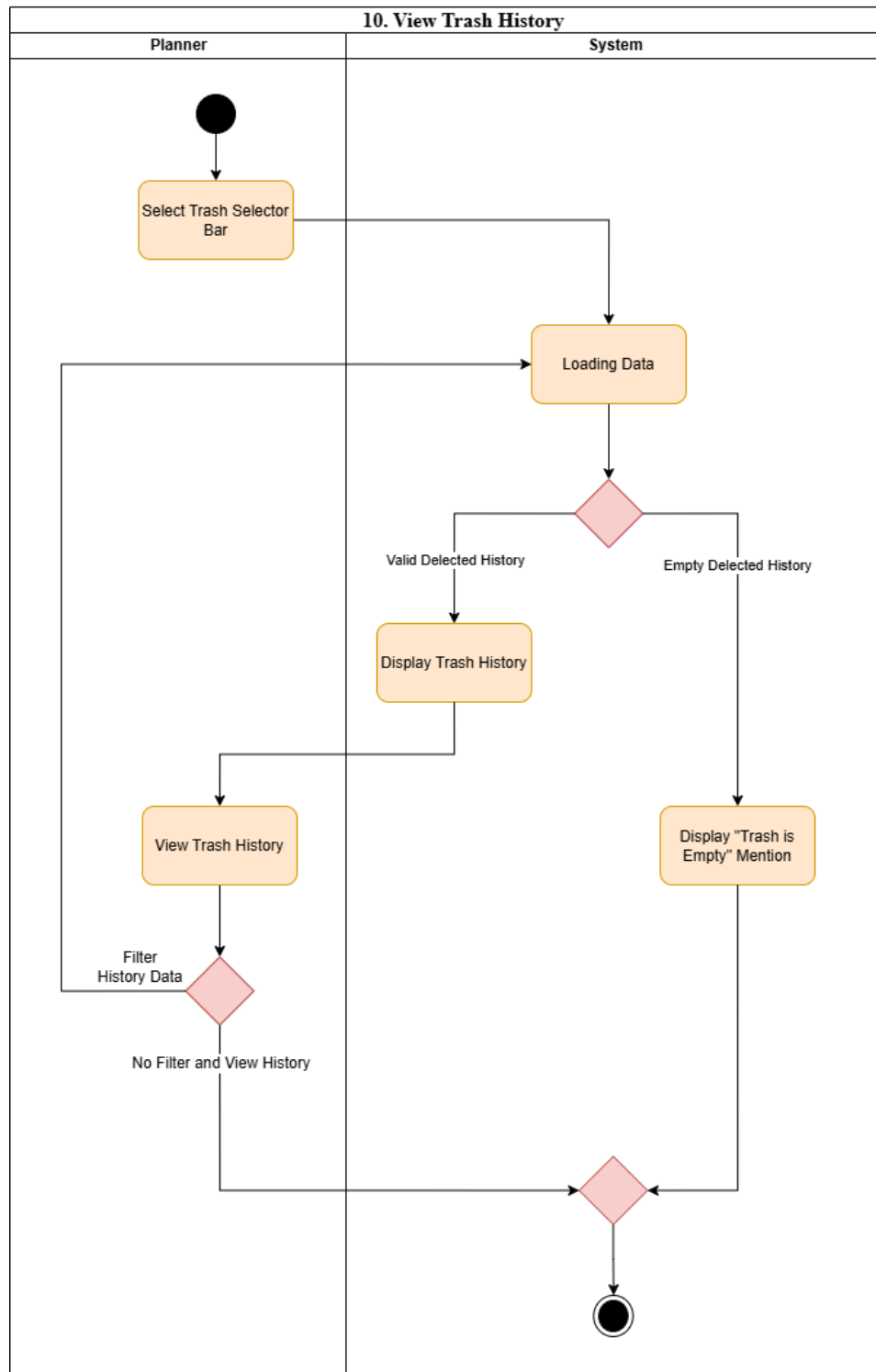


## 8. Searching Projects and Tasks

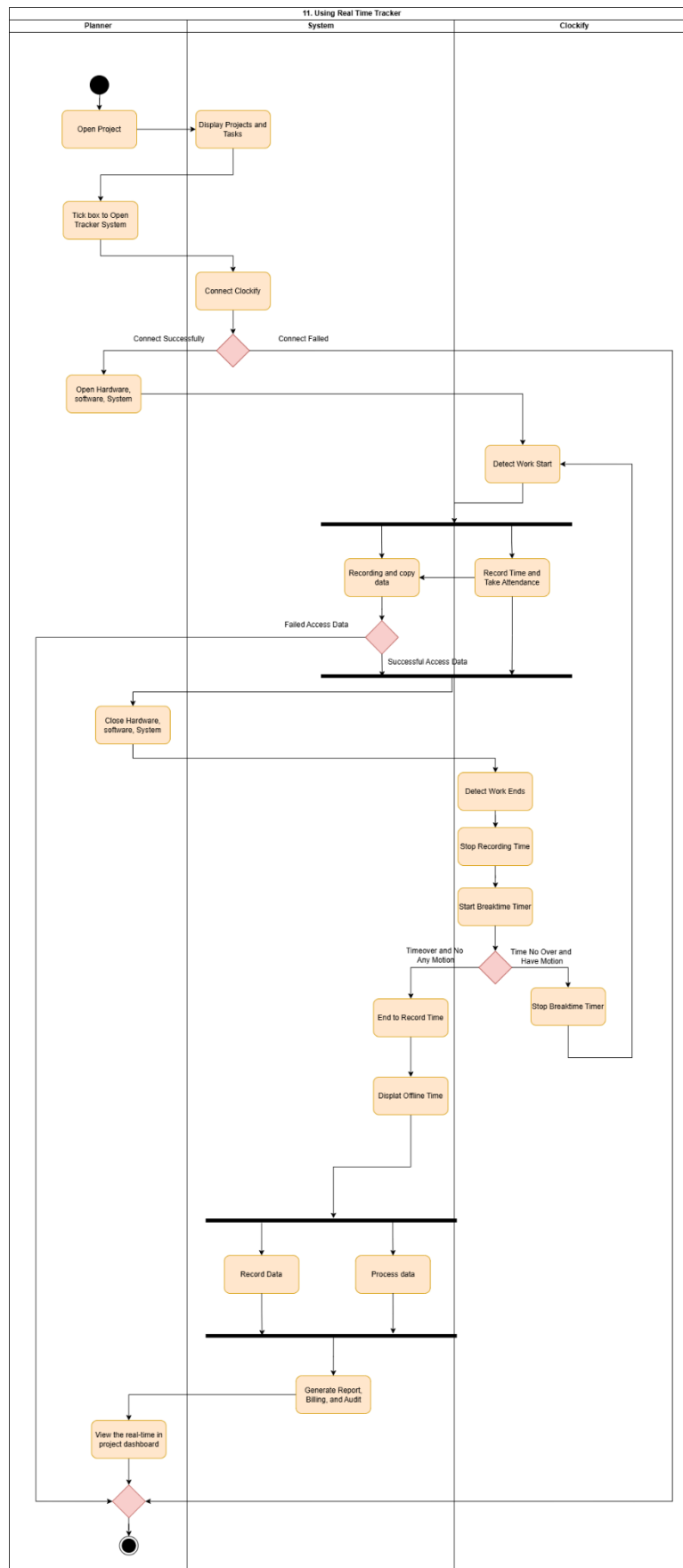


## 9. Sharing Project Dashboard

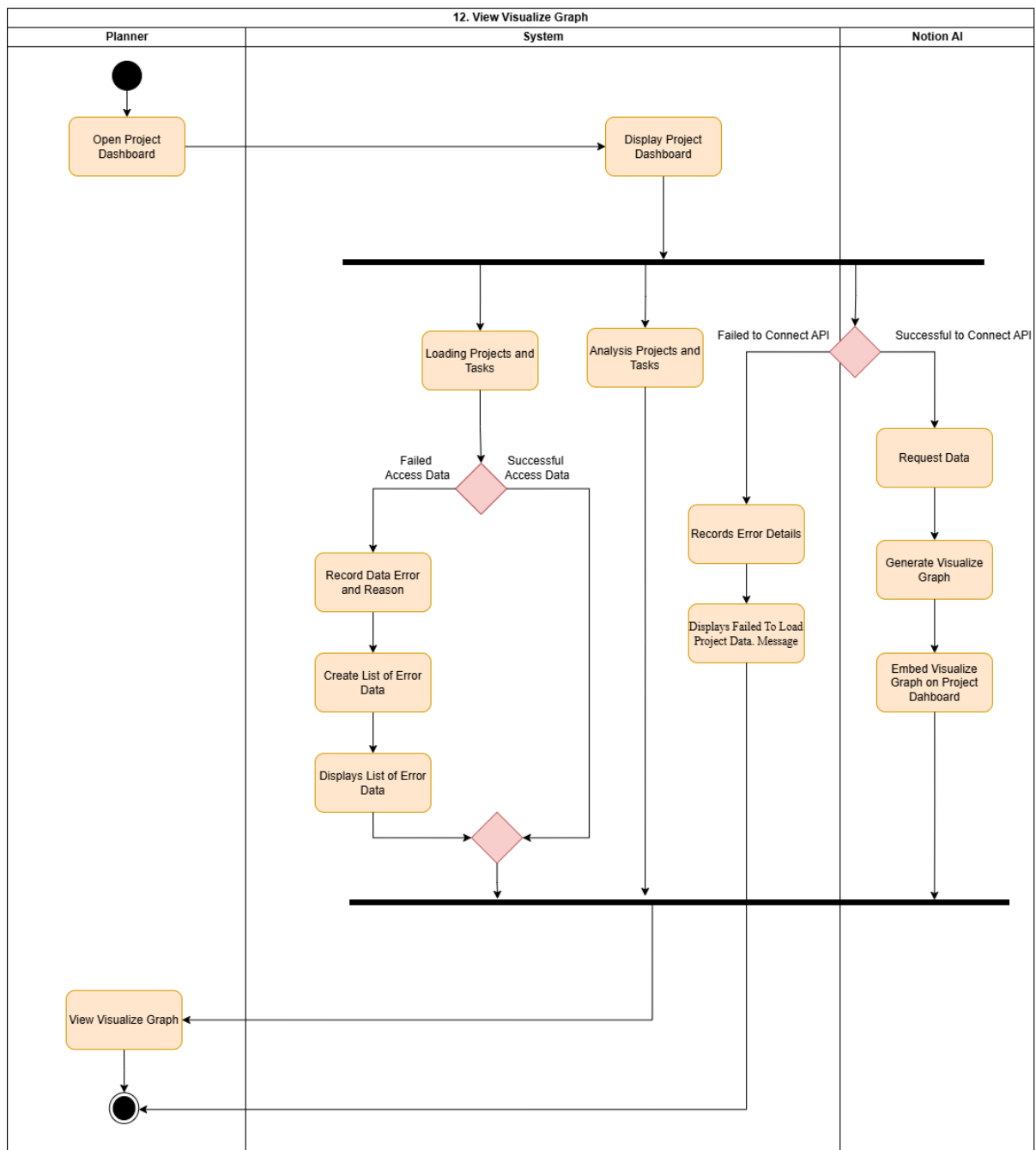


**10. View Trash History**

## 11. View Real Time Tracker



## 12. View Visualize Graph



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